

Vending Machine Demand Forecasting & Inventory Deck

Device & SKU Classification, Model Accuracy Analysis, Safety Stock Optimization, and Strategic Operational Roadmap

 Operations Analytics Team | DE Academy

Executive Summary & Core Goals

🎯 Project Objective

Implement machine learning-driven time series forecasting (XGBoost) across vending device networks to optimize inventory replenishment, eliminate stockouts, and streamline logistics routes.

Transition from reactive replenishment schedules to proactive dynamic reorder triggers based on 7-day rolling sales velocity.

📊 Target Outcomes

18.8% Holdout MAPE: Achieve high forecast accuracy across fast-moving SKUs.

Logistics Consolidation: Reduce restock visit frequency on low-volume devices while preventing stockouts on top-tier devices.

Key Demand & Model Metrics

18.8%

Holdout MAPE

Achieved by XGBoost model, significantly outperforming classical ARIMA baselines.

62%

7d Rolling Importance

Primary predictive feature weight driving overall demand forecast stability.

-0.90

Restock Correlation

Pearson correlation confirming strong historical reactive restock patterns.

Device Fleet Classification

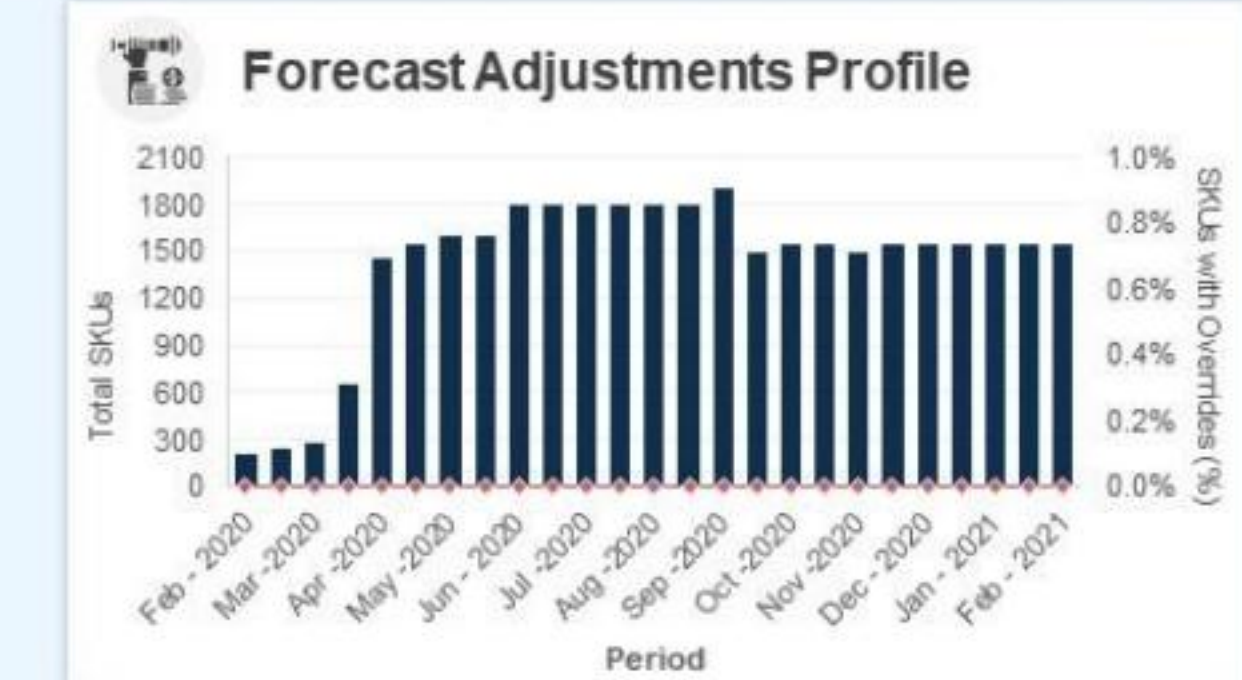
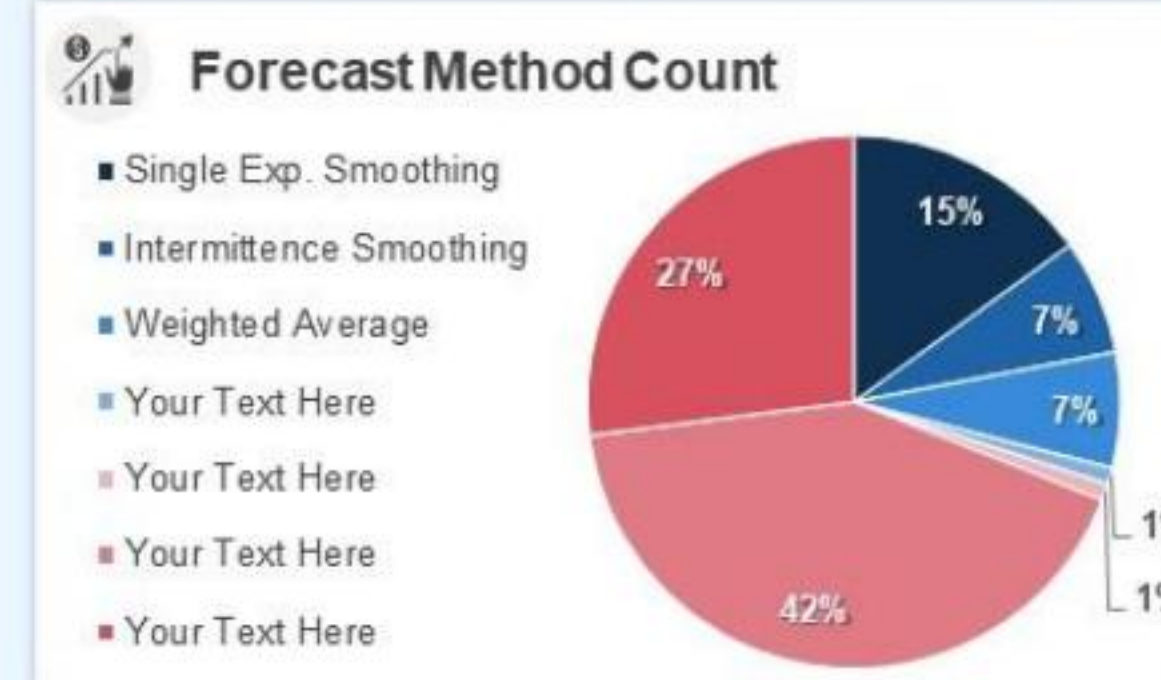
Device Category	Device Count	Volume & Restock Profile	Operational Action Plan
Overused	3 Devices	High Volume / Frequent Restocks	Maintain scheduled 2-day replenishment; optimize batch sizes.
Inefficient	1 Device	Low Volume / Frequent Restocks	Reduce restock visit frequency to consolidate route logistics.
Underused	0 Devices	Low Volume / Infrequent Restocks	Evaluate machine placement and potential fleet reallocation.
Stockout Risk	0 Devices	High Volume / Infrequent Restocks	Fleet healthy; no immediate acute stockout risk flagged.

Predictive Driver Analysis

- **7-Day Rolling Mean Dominance:** Accounts for ~62% of total feature importance in XGBoost, acting as the primary anchor for forecast accuracy.
- **Lag Features (t-1, t-7):** Form the secondary tier of predictive weight, effectively capturing short-term momentum and weekly consumption cycles.
- **Calendar Context Flags:** Features such as *is_weekend* and *quarter* provided minor incremental gains once short-term demand signals were incorporated.

Market Demand and Forecast Dashboard with Method Count

This graph/chart is linked to excel, and changes automatically based on data. Just left click on it and select "edit data".

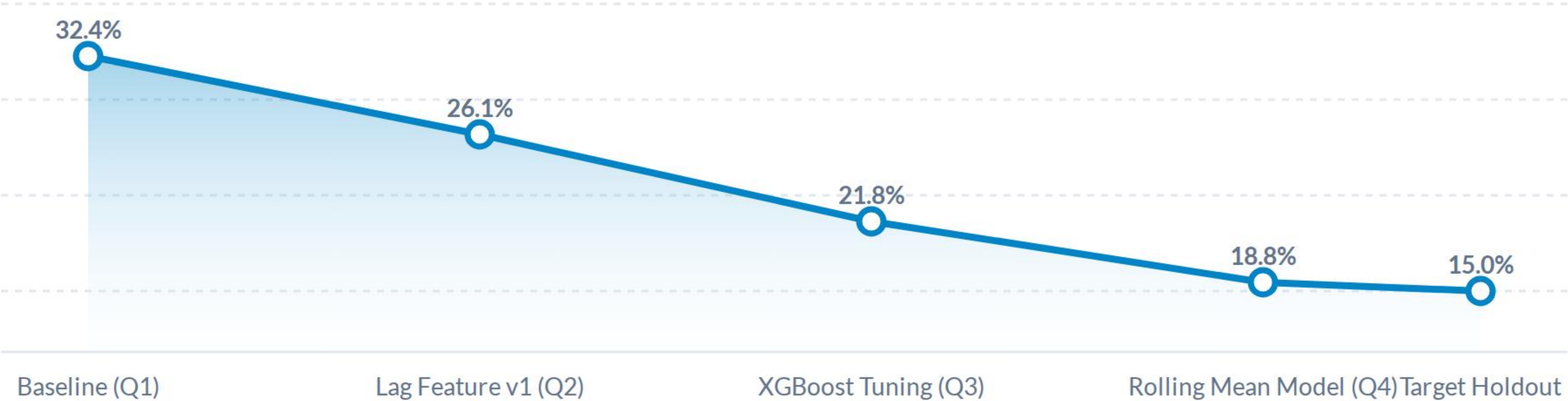


SKU Inventory Optimization

$$ROP = (\bar{D} \times L) + Z \times \sigma_D \times \sqrt{L}$$
 (Service Level = 95%, Z = 1.65, Est. Lead Time L = 2 Days)

SKU Identifier	7d Avg Demand	7d Std Dev	Safety Stock	Reorder Point	Variance Status
da3562c5...fb66aa	19.4	12.4	28.9	67.8	High Volatility
b1cfc8c8...3deb	17.4	7.8	18.3	53.2	Normal
93d664dd...51e92	19.3	4.8	11.2	49.7	Normal
9a154810...b7e05	16.0	7.3	17.1	49.1	Normal
be61be55...88fa7	14.1	6.6	15.4	43.7	Normal

Forecast Accuracy & MAPE Trend



Continuous reduction in MAPE error rate following iterative feature engineering and XGBoost hyperparameter tuning.

Smart Retail & Monitoring

Real-Time Telemetry & Tracking

Deploying automated telemetry across smart vending machines allows real-time inventory monitoring and dynamic anomaly detection.

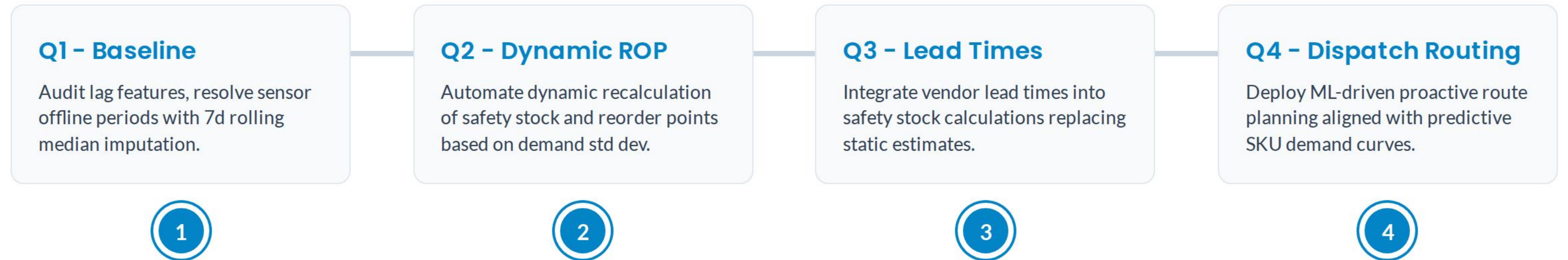
Audited 2.4% missing lag windows caused by maintenance offline periods and applied 7-day rolling median imputation to safeguard time-series input continuity.



Fleet Restock Dispatch Plan

Device ID	Facility Location	Restock Frequency	Optimal Dispatch Window	Target Batch Size	Route Priority
VM-101	Industrial Plant A	2 Days (M / W / F)	Mon / Wed / Fri – 06:00 AM	180 Units	High Priority
VM-102	Corporate HQ Main	3 Days (T / Th)	Tue / Thu – 08:30 AM	120 Units	Normal
VM-103	Transit Hub Terminal B	Daily Service	Everyday – 05:00 AM	250 Units	Urgent
VM-104	Manufacturing Facility C	2 Days (T / F)	Tue / Fri – 07:00 AM	150 Units	High Priority
VM-105	Distribution Annex	Weekly (Mon)	Monday – 10:00 AM	90 Units	Standard

Strategic Operational Roadmap



Key Actionable Next Steps



Investigate Spikes

Follow up with field operations on mid-February demand spikes to verify promotional orders vs telemetry errors.



Recalculate ROP

Implement automated monthly recalculation of safety stock and reorder points using 7-day rolling variance.



Route Integration

Transition from reactive replenishment to ML-driven route dispatch to maximize driver fleet efficiency.

Questions & Discussion

Vending Machine Demand Forecasting & Inventory Optimization Deck

✉ DE Academy Operations Analytics Team

Image Sources



https://www.slideteam.net/media/catalog/product/cache/1280x720/m/a/market_demand_and_forecast_dashboard_with_method_count_slide01.jpg

Source: www.slideteam.net



https://www.digitalmediavending.com/_next/image?url=%2Fwp-content%2Fuploads%2F2026%2F05%2F2026-05-03-transfigure-retail-hero.jpg&w=3840&q=75&dpl=dpl_2TA7zassNdJKGZWdYRdF9ThBKz67

Source: www.digitalmediavending.com